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REVIEW ARTICLE

An expert opinion on heart failure in hypertensive heart disease

Javier Díez [1,*,†], Arantxa González [1,2,3,†], Annet Kirabo [4], Paolo Verdecchia [5], Tazeen H. Jafar[6], Dagfinn Aune [7,8,9], Luke J. Laffin[10], Thilo Burkard [11], Giuseppe M. Rosano[12,13], Massimo Piepoli[14,15], Arthur Mark Richards[16], Begoña López[1,2], Susana Ravassa [1,2], Calvin W.L. Chin[17], Franz H. Messerli [18,19], Thomas H. Marwick[20,21], Miguel Camafort [22,23], Giovanni de Simone [24,25], Jian Zhang[26], Bertram Pitt [27], Marijana Tadic [28], Cesare Cuspidi [29], Faiez Zannad [30], Marco Metra [31], Michael Böhm [32], and Javed Butler [33,34,*]

[1]Program of Cardiovascular Diseases, Center of Applied Medical Research (CIMA) and School of Medicine, University of Navarra, Pamplona, Spain; [2]Center for Biomedical Research in Cardiovascular Diseases Network (CIBERCV), Carlos III Institute of Health, Madrid, Spain; [3]Department of Cardiology and Cardiac Surgery, University of Navarra Clinic, and Navarra Institute for Health Research (IdiSNA), Pamplona, Spain; [4]Department of Medicine, Vanderbilt University Medical Center, Nashville, TN, USA; [5]Associazione Umbra Cuore e Ipertensione and Department of Cardiology, Hospital S. Maria della Misericordia, Perugia, Italy; [6]Program in Health Services & Systems Research, Duke-NUS Medical School, Singapore; [7]Department of Epidemiology and Biostatistics, School of Public Health, Imperial College London, London, UK; [8]Department of Nutrition, Oslo New University College, Oslo, Norway; [9]Department of Research Cancer Registry of Norway, Norwegian Institute of Public Health, Oslo, Norway; [10]Cleveland Clinic Foundation, Cleveland, OH, USA; [11]Medical Outpatient Department and Hypertension Centre, ESH Hypertension Centre of Excellence, University Hospital Basel, Basel, Switzerland; [12]Department of Human Sciences and Promotion of Quality of Life, San Raffaele Open University of Rome, Rome, Italy; [13]Cardiology, IRCCS Raffaele, Cassino, Italy; [14]Clinical Cardiology, IRCCS Policlinico San Donato, Milan, Italy; [15]Department of Biomedical Sciences for Health, University of Milan, Milan, Italy; [16]Christchurch Heart Institute, University of Otago, Christchurch, New Zealand; [17]Department of Cardiology, National Heart Centre Singapore, Singapore, Singapore; [18]Kantonsspital Luzern, University of Bern, Bern, Switzerland; [19]Jagiellonian University, Krakow, Poland; [20]Baker Heart and Diabetes Institute, Melbourne, Australia; [21]Menzies Institute for Medical Research, Hobart, Australia; [22]Hypertension Unit and Heart Failure Unit, Department of Internal Medicine, Hospital Clinic, University of Barcelona, Barcelona, Spain; [23]Center for Biomedical Research in Pathophysiology of Obesity and Nutrition (CIBEROBN), Carlos III Institute of Health, Madrid, Spain; [24]Department of Medicine, The New York Presbyterian Hospital, Weil Cornell Medicine, New York, NY, USA; [25]Department of Advanced Biomedical Sciences, Federico II University, Naples, Italy; [26]State Key Laboratory of Cardiovascular Disease, Heart Failure Center, Fuwai Hospital, National Center for Cardiovascular Diseases, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China; [27]Division of Cardiology, University of Michigan School of Medicine, Ann Arbor, MI, USA; [28]Klinik für Innere Medizin II, Universitätsklinikum Ulm, Ulm, Germany; [29]Department of Medicine and Surgery, University Milano-Bicocca, Milan, Italy; [30]INSERM, Centre d'Investigations Cliniques 1433, CHRU de Nancy, Inserm 1116 and INI-CRCT (Cardiovascular and Renal Clinical Trialists) F-CRIN Network, Université de Lorraine, Nancy, France; [31]Cardiology, Vita-salute San Raffaele University and IRCCS San Raffaele Hospital, Milan, Italy; [32]Klinik für Innere Medizin III and HOMICAREM-Homburg Institute for CardioRenalMetabolic Medicine, Universitätsklinikum des Saarlandes, Saarland University, Homburg/Saar, Germany; [33]Baylor Scott and White Research Institute, Dallas, TX, USA; and [34]University of Mississippi, Jackson, MS, USA

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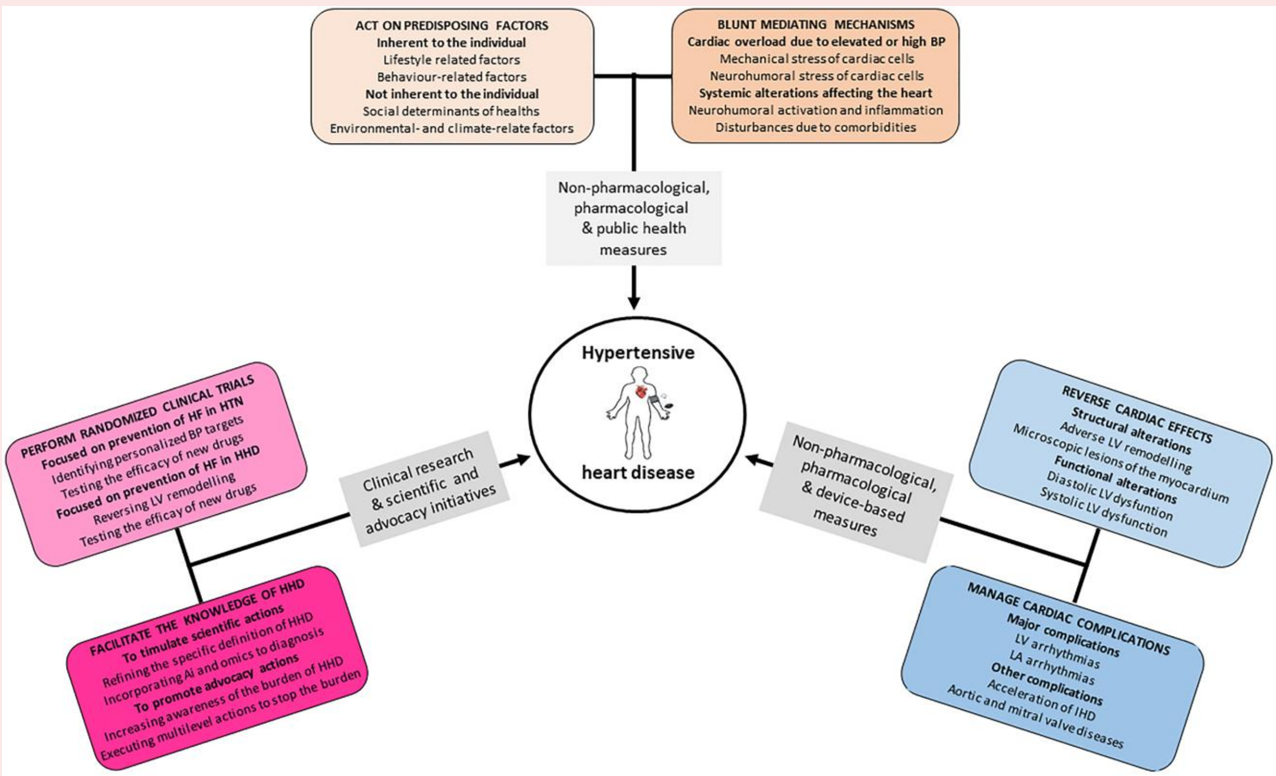
Abstract

There is strong evidence that hypertension is a major risk factor for heart failure (HF). Hypertension contributes to incident HF through direct and indirect effects. Indirect effects are consequences of ischaemic heart disease because hypertension facilitates atherosclerotic obstructive coronary artery disease. The direct effects are straightly related to hypertensive heart disease (HHD). Hypertensive heart disease poses a significant challenge with substantial medical and public health implications. Efforts should be made to recognize and manage HHD in a timely manner and optimize hypertension treatment. Reducing blood pressure (BP) and/or reassessing antihypertensive therapy using traditional or novel approaches can halt or delay progression to HF in patients with HHD and possibly prevent it. However, HHD's importance as a risk factor for overt HF is often

* Corresponding authors. Email: jadimar@unav.es (J.D.); butlzih@gmail.com (J.B.)
† These authors contributed equally to this work.

overlooked in clinical practice. This document aims to summarize the current understanding of the burden of HHD and its risk for incident HF, discuss the mechanisms underlying HHD-related HF onset and progression, consider how diagnostic tools contribute to individualized phenotyping and HF risk stratification of HHD, address how therapeutic measures ameliorating or even preventing structural and functional alterations of HHD, along with BP control influence HHD-associated HF risk.

Graphical abstract



A holistic proposal for heart failure prevention in hypertensive heart disease. The groups of factors to be targeted for effective prevention are described, as well as the types of actions to be implemented. BP, blood pressure; LV, left ventricular; LA, left atrial; IHD, ischaemic heart disease; HTN, hypertension.

Keywords Adverse left ventricular remodelling • Arterial hypertension • Hypertensive heart disease • Diagnostic and prognostic phenotyping • Antihypertensive treatment • Primary prevention

Introduction

Data collated by the NCD Risk Factor Collaboration (NCD-RisC) show that the number of people with hypertension doubled from 1990 to 2019, from 648 million individuals in 1990 to 1278 million individuals in 2019.[1] On the other hand, data from the Global Burden of Disease (GBD) study 2019 indicate the worldwide prevalence of heart failure (HF) increased by 106.3% between 1990 and 2019.[2] The global prevalence of HF in 2019 has been estimated to be of 3% of the total population, equating to an estimated 56.2 million individuals.[3] Development of HF is one of the main consequences of hypertension.[4] Reducing the risk of HF due to hypertension will not be possible unless the challenges posed by hypertension-mediated cardiac damage [i.e. hypertensive heart disease (HHD)] are effectively addressed. Of note, from 1990 to 2019, the global prevalence of HHD cases increased by 138%, reaching more than 18.5 million individuals according to the GBD.[5]

Although the definition of HHD is a matter of debate, we define it as a condition resulting from chronic hypertension, identifiable by structural and functional changes in the heart, particularly disturbed left ventricular (LV) geometry, and LV diastolic and/or systolic dysfunction. It leads to HF not primarily depending on coronary artery disease (CAD), often with preserved LV ejection fraction (HFpEF). This definition does not exclude other frequently co-existing anomalies affecting the left atrium and the right heart.[6,7]

Comprehensive epidemiological studies show that HHD is one of the two leading causes of HF in men and women worldwide, along with ischaemic heart disease (IHD).[2,8,9] Hypertensive heart disease represents a significant challenge in terms of HF burden, and deserves considerations beyond the general European and American guidelines on arterial hypertension[10–12] or HF.[13,14] Therefore, the clinical and public health relevance of HHD deserves to be highlighted.

Accordingly, this document summarizes the current understanding of the disease burden of HHD and its association with HF, discusses present knowledge about the mechanisms involved in the onset and progression of HHD and evaluates diagnostic and phenotyping procedures for HHD and their added prognostic value. It also reassesses therapeutic interventions and healthcare policies aimed to prevent HF in HHD, and discusses uncertainties and gaps about HHD that merit future research.

Epidemiological impact of hypertension and hypertensive heart disease on heart failure risk

Hypertension as a general risk factor for heart failure

Taking into account published data from the ESC Long-Term Heart Failure Registry, hypertension (treated or untreated) was considered the primary aetiology of HF in 66.4% of patients.[15] Among HF with reduced ejection fraction (HFrEF) patients, 4.5% presented untreated hypertension and 55.6% were treated, whereas among HF patients with preserved EF (HFpEF) 18.1% were untreated and 67% were treated. Regarding HF with mid-range LV ejection fraction, 9.6% were untreated hypertensives and 60.1 were treated.[15] Similar findings have been reported by other well-characterized cohorts.[16]

Hypertensive heart disease as specific cause of heart failure

According to data from the GBD 2019 study, which estimated key health metrics across 204 countries with disease classified following the International Classification of Diseases (ICD) codes, in 1990 and 2019, HHD accounted for 43.6% and 41.1% of all contributing causes of HF, respectively (*Figure 1*).[2] In the same years IHD accounted from for 29.0% to and 33.5%, respectively (*Figure 1*).[2] In 2019, HHD and IHD accounted for 33.2% and 37.7% of severe HF cases, respectively (*Figure 1*).[2] However, it cannot be excluded that concurrent coding was used when evaluating contributing causes to HF. In this context, if concurrent conditions are present, HHD should not be designated as the primary cause of HF, although it undoubtedly contributes to its onset and progression. On the other hand, there are no robust data defining the severity or duration of hypertension required for HF development, and further studies are needed.

The examination of the contributing causes of HF across socio-demographic index (SDI) quintiles shows that HHD and IHD were more prevalent in higher SDI countries than in lower SDI countries, although limitations in data availability from low-income countries may affect this assessment.[2] As recently reported,[17] the proportion of women with HF due to HHD is higher compared with men, in whom the predominant cause of HF is IHD. The proportion of age-standardized prevalence rate of HF due to each cause varied widely by age group in 2017.[18] In adults aged 25–69 years, HHD was the most frequent cause of HF.[18]

The growing and threatening epidemiological relevance of HHD as a cause of HF is highlighted by a recent study of comprehensive forecasts of future trends in disease burden in 204 countries and territories to 2050, which shows that the age-standardized disease burden will continue to decline in the future with three notable cause-specific exceptions, including HHD, diabetes mellitus (DM), and chronic kidney disease (CKD).[19]

Pathophysiology of heart failure due to hypertensive heart disease

Adverse left ventricular remodelling

Hypertensive adverse LV remodelling is the result of a dynamic process in which exposure to haemodynamic overload alters LV anatomy and function. The balance between pressure and volume overload in arterial hypertension results in a range of alterations. The pressure overload imposed by hypertension increases LV pressure, which influences LV relaxation, leading to Grade I LV diastolic dysfunction as the initial alteration in HHD (*Figure 2*).[20] Increased LV pressure also increases LV wall tension which triggers a compensatory response, inducing inward LV wall thickening [resulting in concentric LV remodelling or mostly in concentric LV hypertrophy (LVH)] (*Figure 2*).[20] When LV pressure overload is chronic, LV diastolic dysfunction worsens, concentric LVH is not sufficient to compensate for the maintenance of LV wall tension, LV performance decreases (i.e. low stroke volume and cardiac output in the presence of normal systolic function) and HFpEF develops (*Figure 2*).[20]

Although HFpEF is the most common type of HF seen in patients with HHD, there is a possibility that long-standing hypertension may lead also to LV systolic dysfunction, which could contribute to development HFrEF.[20] In this regard, it is worth noting that pressure overload can affect mechanical deformation of myocardial fibres during the cardiac cycle (strain), which plays a key role in affecting LV contractility and systolic function. In fact, global longitudinal strain (GLS) impairment occurs early in the non-hypertrophied, pressure-overloaded LV in hypertension,[21] which can be a precursor to LV ejection fraction (LVEF) deterioration,[22] and a risk factor for HFrEF.[23] However, because chronic hypertension is associated with and can cause atherosclerotic CAD[24] and structural and functional alterations of coronary microcirculation[25,26] progression from HHD to HFrEF is more frequently, if not exclusively, observed in patients with intercurrent coronary syndromes or chronic subclinical ischaemia.[24,27,28]

Some hypertensive patients may present a predominant volume overload (e.g. patients with obesity and/or CKD). In these cases, adverse LV remodelling consists of eccentric LVH characterized by outward thickening of the LV wall and enhanced chamber diameter.[20] When volume overload is sustained, the eccentric remodelled LV decompensates, and LV dilatation with HFrEF may ensue.[20]

Left atrial (LA) structural and functional alterations commonly parallel adverse LV remodelling in HHD, with a prevalence of LA enlargement that may be >50%.[29–31] Left atrial enlargement in hypertension likely reflects an adaptation of the atrial chamber to a pressure overload primarily driven by increased LV pressure. Left atrial structural alterations show prognostic value for a wide variety of clinical outcomes in HHD patients, including aggravation of LV diastolic dysfunction by impaired atrial contribution to LV diastolic filling,[20] and increased risk of incidence and progression of atrial fibrillation (AF),[32–36] which often precipitates HF.[37]

Histological and molecular changes in the myocardium

In the LV hypertensive myocardium various histological lesions develop (*Figure 3*). They include, cardiomyocyte hypertrophy (i.e. due to transversal growth by parallel addition of sarcomeres or due to longitudinal growth by in series addition of sarcomeres), fibrosis [i.e. reactive interstitial and perivascular fibrosis, and small foci of reparative fibrosis (microscars)], and alterations of coronary microcirculation (i.e. decreased lumen-to-wall ratio of prearterioles 100–500 μm diameter and arterioles <100 μm diameter, and capillary rarefaction).[25,26] While fibrosis is due to the deposition of an excess of collagen Type I and III fibres, the pre-arteriolar and arteriolar geometric changes are due to hyperplasia or realignment of vascular smooth muscle cells (VSMCs) in the vascular wall, and the reduction in capillary density is probably related to alterations in pericytes surrounding capillary endothelial cells.[25,26]

The importance of the myocardial changes is increasingly recognized, as they critically contribute to altering LV and LA function and electrical activity. Fibrosis is associated with increased LV stiffness, reduced myocardial strain, LV diastolic and systolic dysfunction, and LV and LA arrhythmias (namely AF) in patients with HHD.[38] Also coronary microvascular dysfunction contributes to LV dysfunction and LV arrhythmias in patients with HHD without CAD.[39]

Histological lesions of the hypertensive myocardium are accompanied by several molecular alterations. Beyond the re-expression of foetal

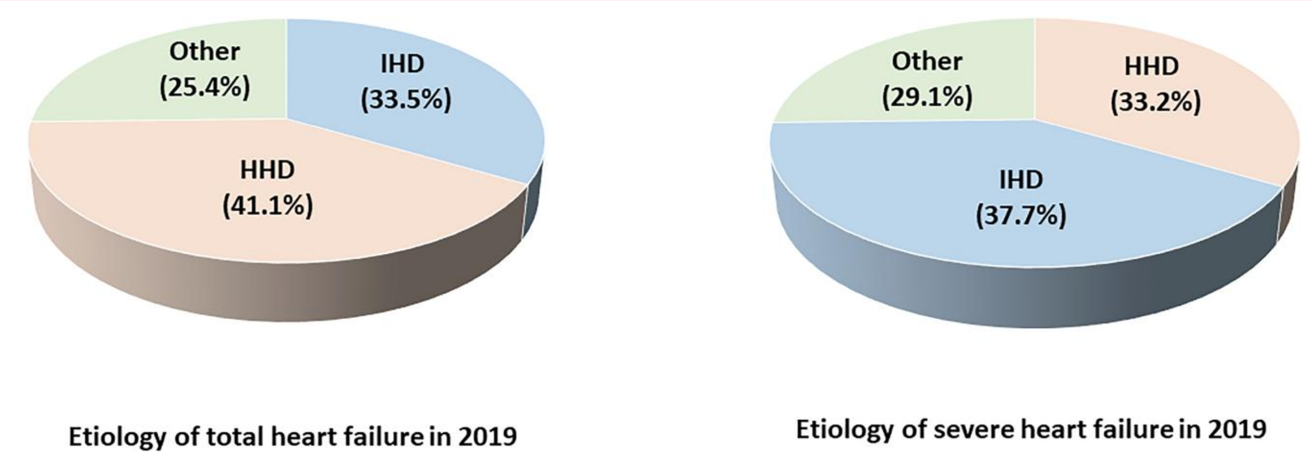


Figure 1 Causes of total (left panel) and severe (right panel) heart failure in 2019 according to global data from the Global Burden of Disease study. IHD, ischaemic heart disease; HHD, hypertensive heart disease; Other, other causes different from IHD and HHD. Prepared from data in reference[2]

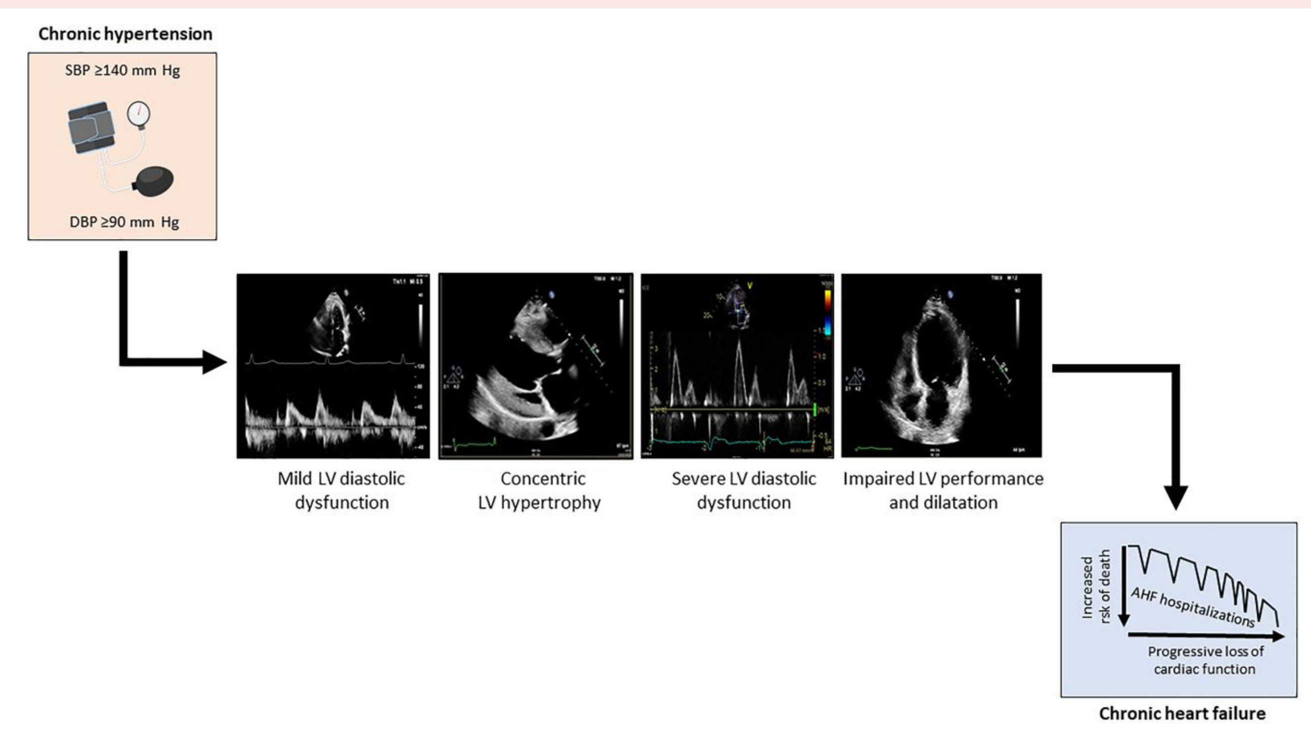


Figure 2 Pathophysiological continuum of hypertensive heart disease, which describes the progressive process of adverse remodelling of the left ventricle that is initiated by chronic exposure to hypertension and finally manifest as chronic heart failure. Echocardiographic images are courtesy of Dr Agnes Díaz, Clinica Universidad de Navarra, Pamplona, Spain

genes leading to cellular hypertrophy, there are other molecular alterations in hypertrophic cardiomyocytes. These include reduction in mitochondrial mass and density accompanied by deterioration of mitochondrial metabolic and bioenergetic functions, and inhibition of survival pathways facilitating the loss of cardiomyocytes via apoptosis.[40,41] In addition hypertrophic cardiomyocytes also exhibit abnormal expression of ion channels and/or junctional complexes that prolong the duration of the action potential and affect the refractoriness of the cell. These alterations make the myocardium electrically vulnerable[42] and explain why patients with LVH have a threefold higher risk of ventricular tachycardia[43] and sudden arrhythmic death[44] than hypertensive patients without HHD.

Cardiac fibroblasts undergo[45] These changes activate differentiation in myofibroblasts, mostly mostly driven by transforming growth factor-β, and the subsequent production of a pro-fibrotic secretome that leads to alterations in the extracellular turnover of fibrillar collagen

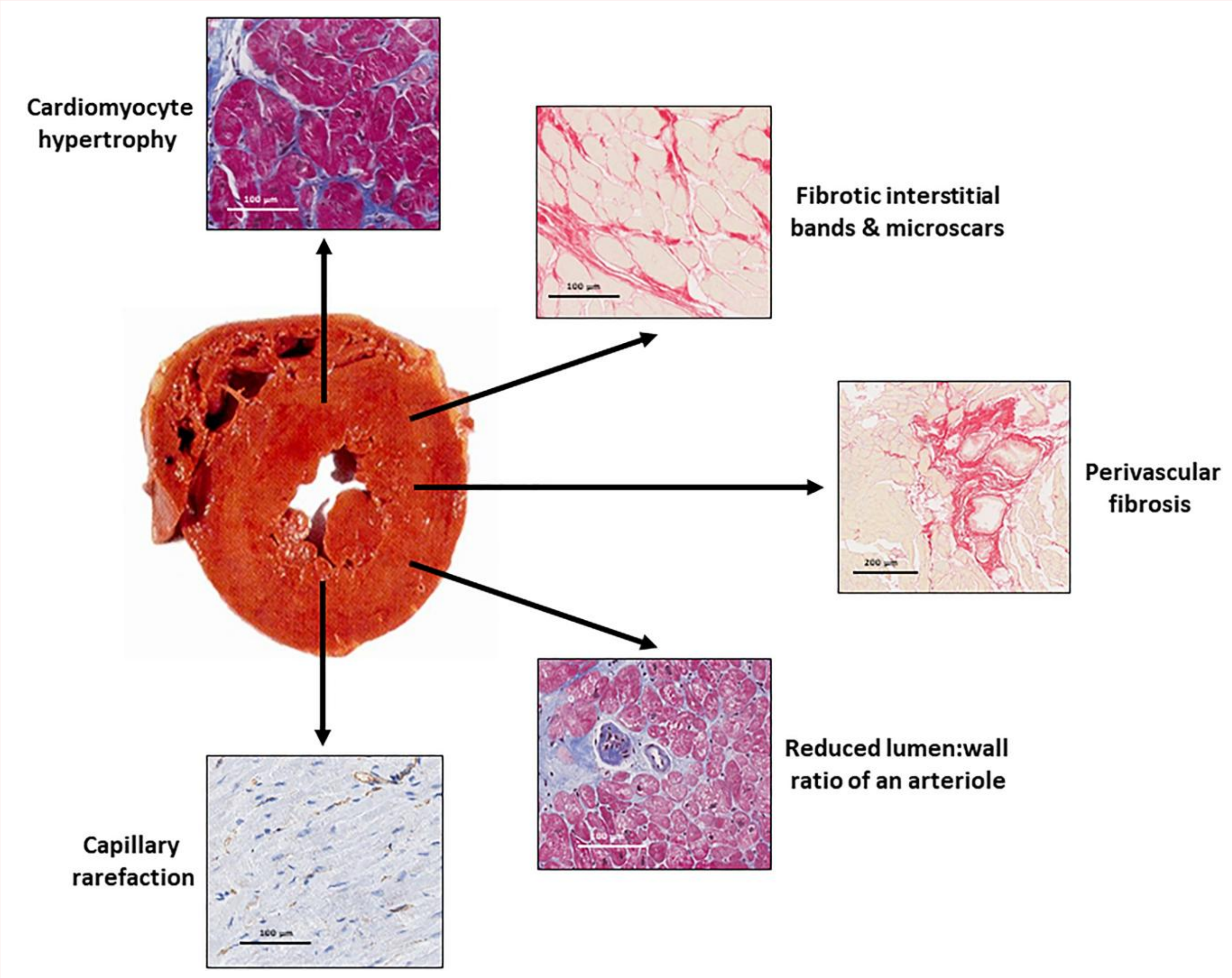


Figure 3 Main microscopic lesions found in the myocardium of a hypertensive patient with concentric left ventricular hypertrophy: cardiomyocyte hypertrophy (Masson's trichrome staining), interstitial and perivascular fibrosis (picrosirius red staining), reduced lumen:wall ratio of an intramyocardial arteriole (Masson's trichrome staining), and capillary rarefaction (von Willebrand staining). Microscopic images are courtesy of Dr Begoña López, Center of Applied Medical Research, Pamplona, Spain

Types I and III.[46] These alterations favour the predominance of collagen synthesis over collagen degradation and its excessive deposition.[46]

The presence of low-grade inflammation of both the vascular wall and perivascular fat, in conjunction with combined with the effects of vasoactive agents such as angiotensin II and endothelin-1, and alterations in the interaction with the extracellular matrix of the vascular wall, contributes to the growth and realignment of VSMCs in prearterioles and arterioles in hypertension.[47] Conversely, heightened production of reactive oxygen species and diminished bioavailability of nitric oxide in the endothelial cells of the coronary microcirculation in HHD result in vasoactive dysfunction and pro-inflammatory activation.[47]

Effects of comorbidities

In addition to hypertension, other concomitant conditions may contribute to the development and aggravation of HHD, especially obesity. Both normotensive and hypertensive groups, obese persons exhibit significantly higher values of indexed LV mass (LV mass/height$^{2.7}$) or LV mass index (LVMI) than normal-weight individuals, and that this difference is independent of blood pressure (BP) values.[48] Obesity-related increased plasma volume, neurohormonal activation and adipokine-mediated inflammation and oxidative stress further exacerbate the impact of hypertension on the LV resulting in higher LV filling pressure and enhanced myocardial fibrosis which, in turn, aggravate LV dysfunction and facilitate HFpEF.[45,49,50]

A cardiac magnetic resonance (CMR) study of asymptomatic hypertensive patients with and without DM and free of CAD suggested the presence of more severe concentric LVH, worse LV function and a higher fibrosis burden in patients with hypertension and DM compared with patients with hypertension alone.[51] Furthermore, in hypertensive patients the coexistence of both DM and metabolic syndrome is associated with the highest prevalence of concentric LVH and the most severe reduction in myocardial mechano-energetic efficiency.[52]

When associated with CKD, hypertension-mediated damage on the heart and other organs increases. Chronic kidney disease worsens the myocardial lesions, clinical state and outcomes in HHD through several pathways (e.g. volume overload, sympathetic overactivity, calcium metabolism disorders) leading to an excess of parathyroid hormone and fibroblast growth factor-23, over-synthesis of pro-inflammatory cytokines, and retention of uraemic solutes.[53]

Cardiac imaging in the diagnosis of hypertensive heart disease

Phenotypic and prognostic implications

A study on the clinical and pathological characteristics of HHD in victims of sudden cardiac death from a national registry of cardiovascular (CV) pathology in the UK, of all the patients who met clinical and pathological diagnostic criteria for HHD, only 2.7% had an antemortem diagnosis,[54] underscoring the challenges of HHD phenotypic characterization and clinical risk stratification.

Although performing an electrocardiogram (EKG) is part of the recommended initial evaluation of hypertension,[10,11] its suboptimal accuracy in detecting LVH, makes the use of alternative multimodal imaging (accompanied by the assessment of some circulating biomarkers) necessary to confirm the diagnosis of HHD and, even more, its phenotype and related prognosis.[55] The different imaging modalities have specific indications in the evaluation of the heart in hypertensive patients, particularly those with HHD (*Figure 4*).

Echocardiography

In clinical practice, echocardiography is useful to recognize the most important anatomical and functional features of HHD in many patients,[55] and to confirm the presence of LVH in cases that might change their risk stratification, based on EKG.[56]

A comprehensive 2D echocardiographic examination should include morphological parameters for the calculation of LVM which is generally indexed to body surface area or height$^{2.7}$ (more accurate in the presence of obesity) to obtain LVMI.[57] The additional computation of relative wall thickness (RWT) defines at least the three main phenotypes of LV growth in HHD [i.e. concentric remodelling (normal LVMI and increased RWT), concentric LVH (increased LVMI and RWT), and eccentric LVH (increased LVMI and normal RWT)]. More recently, a four-group classification of LV geometry includes the afore mentioned three categories plus concentric LVH with dilatation, with important haemodynamic implications.[58]

Two-dimensional echocardiography also measures LV end-diastolic and end-systolic volumes (EDV and ESV, respectively), LVEF, and LA volume.[57] Tissue-Doppler echocardiographic examination evaluates more sensitive global and regional indices of LV diastolic and systolic function.[57] Global longitudinal strain based on speckle tracking echocardiography (STE) evaluates LV longitudinal deformation as a quantitative assessment of global LV systolic function, less dependent on loading conditions than more traditional measures.[57] Of note, STE can be also used to measure LA strain and evaluate LA dysfunction in different phases (reservoir, conduit, and contraction). Finally, in newly diagnosed, never-treated hypertensive patients, 3D echocardiography-derived LVM/EDV ratio identifies a higher prevalence of LV concentric geometry than 2D echocardiography-derived RWT.[59] Stroke volume is independently and negatively associated with LVM/EDV ratio and its reduction represents an early marker of LV dysfunction in hypertensives with LV concentric geometry.[59]

From a prognostic point of view, there is a strong, continuous, and independent association between LVMI and CV risk in hypertensive patients, independently of BP.[60] Left ventricular diastolic function deteriorates in parallel with worsening LV geometry,[61] and the presence of LV diastolic dysfunction is a potential precursor of HFpEF in patients with hypertension.[62,63]

Cardiac magnetic resonance and other imaging techniques

Cardiac magnetic resonance is considered the most accurate technique for the evaluation of LVM and cardiac chamber volumes, and its measurements do not depend on the geometry of the heart.[64,65] Therefore, CMR-defined LV geometric patterns can have prognostic value in HHD as shown in the general population, with low risk of incident HF or CV death in patients without LVH or with isolated eccentric LVH, a significantly higher risk in patients with isolated concentric LVH, and the highest risk in patients with either concentric or eccentric LVH and LV dilatation (*Figure 5*).[66]

The main advantage of CMR is that it can provide information on tissue characteristics, in particular of the presence and severity of myocardial fibrosis, using late gadolinium enhancement (LGE) for assessing small foci of non-ischaemic reparative fibrosis, and T1 mapping with the evaluation of myocardial extracellular volume (ECV) for assessing interstitial reactive fibrosis.[64,65] In patients with HHD the increase in CMR-assessed ECV is associated with LVH, LV diastolic dysfunction and impairment of GLS.[67–72] Non-ischaemic LGE is also associated with LV diastolic dysfunction and altered myocardial strain in HHD.[72–74] However, standardized CMR measures are needed to establish cut-off values with clinical usefulness for the assessment of fibrosis.[75]

Cardiac magnetic resonance can be used to differentiate HHD from other cardiomyopathies that also present with LVH (e.g. aortic stenosis, hypertrophic cardiomyopathy, Fabry's disease, amyloidosis, and sarcoidosis).[64,65] This is done on the basis of morphological patterns and location of LVH, myocardial tissue characterization, and functional parameters.

In a recent systematic review and meta-analysis computed tomography (CT)-derived ECV showed excellent correlation with CMR-derived ECV.[76] However, the presence of hypertensive patients in the included studies was low, and larger prospective studies are needed to examine the accuracy and diagnostic and prognostic utility of CT to address myocardial fibrosis in HHD.

Imaging techniques may be helpful to evaluate coronary microvascular dysfunction in HHD. Patients with HHD and patients with HFpEF free of CAD have abnormally reduced coronary flow reserve (CFR), assessed by CMR phase contrast imaging.[77] Coronary flow reserve was reduced in patients with HHD when compared with control subjects, but the reduction was greater in patients with HFpEF.[77] In hypertensive patients without CAD and HFrEF, those with LVH had lower CFR on positron emission tomography myocardial perfusion imaging than those without LVH.[78] Reduced CFR in patients with LVH was independently associated with LV diastolic dysfunction, reduced GLS and higher incidence of HF hospitalization and death.[78] These data confirm the relation between coronary microvascular dysfunction, global LV dysfunction and risk of HF.

Circulating biomarkers in the diagnosis of hypertensive heart disease

Biomarkers related to stress-injured cardiomyocytes

Observational data demonstrate that LVH accompanied by biomarker evidence of chronic cardiomyocyte stress and injury, as measured by increased plasma levels of N-terminal pro-B-type natriuretic peptide (NT-proBNP), and high-sensitivity cardiac troponin T (hs-cTnT), identifies a malignant LVH phenotype that is at particularly high risk for HF and death.[79–81] For example compared with asymptomatic patients without EKG LVH and normal levels of the two biomarkers

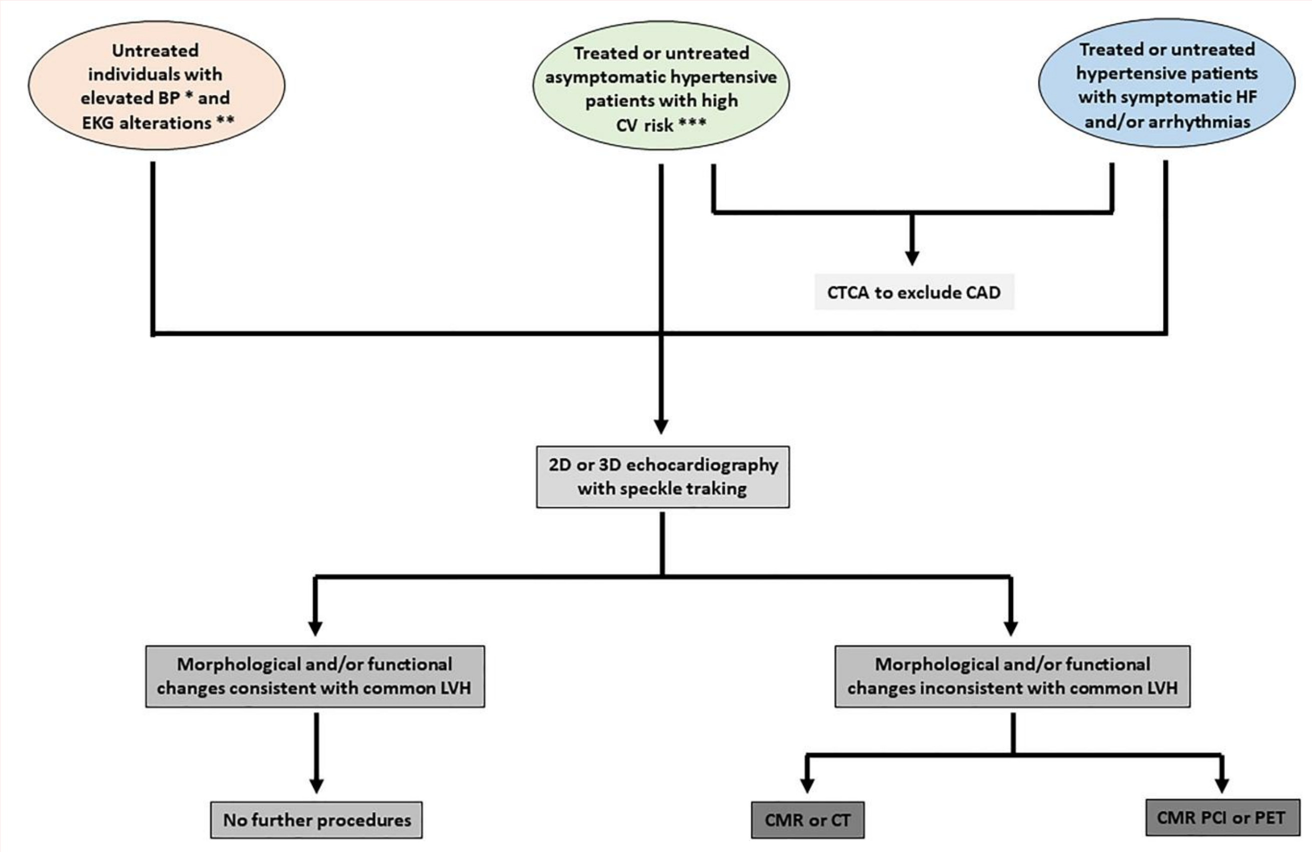


Figure 4 Proposed sequential use of multimodal cardiac imaging in hypertensive patients. BP, blood pressure; EKG, electrocardiogram; LVH, left ventricular hypertrophy; CTCA, computed tomography coronary angiography; CMR, cardiac magnetic resonance; CT, computed tomography; CMR PCI, CMR phase contrast imaging; PET, positron emission tomography. *As defined in references[10–12]. **Electrocardiogram alterations include left ventricular hypertrophy and subclinical arrhythmias. ***Defined by the presence of comorbidities, presence of extra-cardiac hypertension-mediated organ damage and electrocardiogram alterations as defined before

(NT-proBNP <125 pg/ml and cTnT < 14 pg/ml), asymptomatic patients with EKG LVH and elevated biomarkers had a four-fold higher and a two-fold higher risk of HF and all-cause mortality, respectively.[81] The same study found that intensive reduction of systolic BP (<120 mmHg) not only prevented malignant LVH, but can also provide large absolute risk reductions for HF events and death when malignant LVH is present.[81]

The 2024 European Society of Cardiology guidelines proposes use of biomarkers for CV risk stratification if doubts persist after a full screening of CV risk with level B of evidence and Class IIb recommendation.[11] However, the 2023 European Society of Hypertension guidelines suggests that measurement of NT-proBNP and cTnT can be extended to primary care patients, thus allowing useful patient phenotyping.[10] Further studies are needed to assess whether measuring biomarkers related to stress-injured cardiomyocytes will be useful in identifying those patients with HHD who are more likely to develop HF in the future and who, in turn, merit more intensive antihypertensive treatment.

Biomarkers related to myocardial fibrosis

Since myocardial fibrosis impairs LV and LA function, in addition to facilitating arrhythmias, and thus influences the evolution and outcome of HHD, its assessment is clinically relevant. Beyond the use of CMR for the diagnosis of myocardial fibrosis, several circulating biomarkers have been proposed for the non-invasive assessment of this lesion, although only a few of them truly reflect the accumulation of fibrous tissue in the myocardium of the hypertensive heart.

Serum concentrations of the carboxy-terminal propeptide of procollagen Type I or PICP (cleaved from the secreted procollagen Type I molecule during its conversion into the fibril-forming mature collagen Type I molecule by the enzyme procollagen C proteinase) are abnormally increased in patients with HHD.[82] This increase is even more prominent in patients presenting both HHD and HF.[83] Of note, in these studies circulating PICP was directly correlated with the amount of myocardial collagen assessed in endomyocardial biopsies (EMBs), independently of BP.[82,83] Therefore, serum PICP may be an indirect biomarker of the quantity of collagen Type I-dependent myocardial fibrosis in HHD.

Increased cross-linking of collagen Type I fibrils by the enzyme lysyl oxidase makes resulting collagen Type I fibres stiffer and more resistant to degradation by matrix metalloproteinase-1 (MMP-1).[84,85] Therefore, in highly cross-linked collagen Type I fibrils the release of the carboxy-terminal telopeptide of collagen Type I (CITP) by MMP-1 is diminished.[84,85] Indeed, serum CITP to MMP-1 ratio (CITP:MMP-1 ratio) is inversely correlated with myocardial collagen cross-linking in EMBs,[86] which, in turn, is associated with non-invasively assessed LV chamber stiffness[87] in patients with HHD and HF, suggesting that it may be an indirect circulating biomarker of the stiffness of collagen Type I fibres deposited in the myocardium of this patient group.

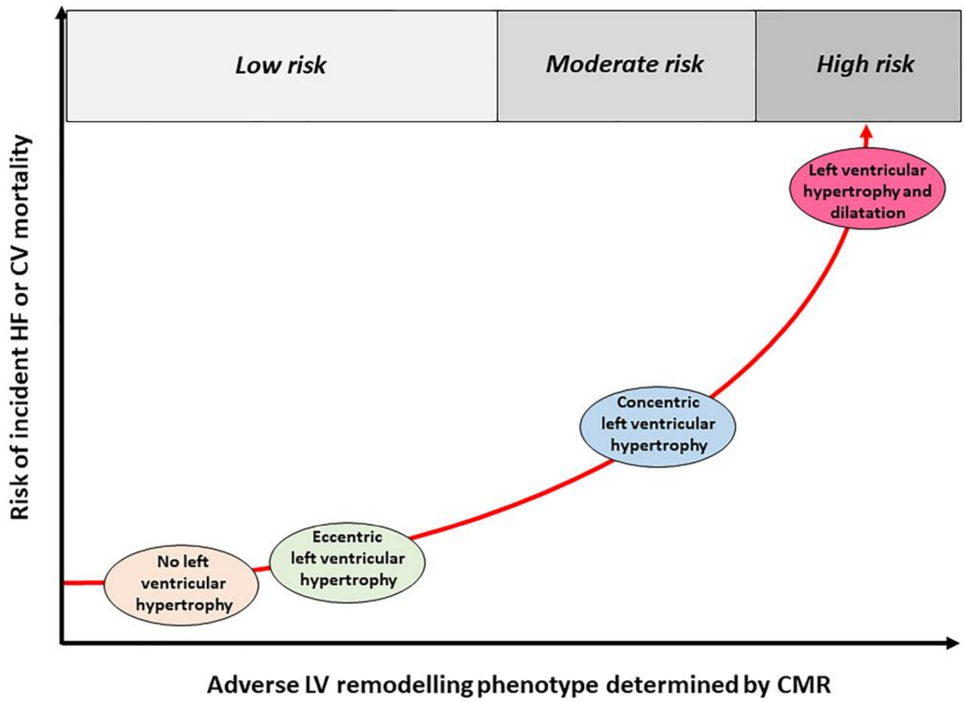


Figure 5 Proposed non-linear relationship between adverse left ventricular remodelling phenotype determined by cardiac magnetic resonance and the risk of heart failure or cardiovascular mortality in hypertensive and non-hypertensive patients without heart failure in the general population. See text for explanations. Prepared from data in reference[66]

Notably, hypertensive patients with HF with a circulating bioprofile of high PICP and low CITP:MMP-1 ratio, reflecting increased myocardial deposition of highly cross-linked collagen type I fibres, present a higher risk of HF hospitalizations or death than patients (*Figure 6*).[88]

Proposal for a sequential clinical diagnosis of hypertensive heart disease

We propose that the progression of HHD in a hypertensive patient (untreated or treated, and in this case with either controlled or uncontrolled BP) with no other demonstrable heart disease has three consecutive stages (*Table 1*). First, hypertensive patient without HHD: the patient is asymptomatic and has no demonstrable abnormalities on cardiac imaging or circulating biomarkers. Second, patient with asymptomatic HHD: the patient is asymptomatic but has abnormalities on cardiac imaging (growth and/or dysfunction of the LV and/or left atrium) without or with alterations in circulating biomarkers (elevated NT-proBNP and/or elevated cTnT). These patients may present arrhythmias, including silent AF, defined by the presence of subclinical episodes of irregular heartbeats that do not cause any noticeable symptoms, such as palpitations, difficulty breathing, or dizziness. Third, patient with symptomatic HHD: the patient is symptomatic (with signs and symptoms of HF) and has both abnormalities on cardiac imaging and alterations in circulating biomarkers. Additionally, AF or other arrhythmias are frequently present.

Hypertensive patients without HHD could be considered as having risk of HF and patients with asymptomatic HHD could be considered as having pre-HF. Identifying these stages is important for considering treatments to halt or delay the progression from hypertension to overt HF in patients with HHD.

Prevention of heart failure in hypertension and hypertensive heart disease

Prevention in hypertension

Preventing the risk of HF in hypertensive patients, and more particularly in those with HHD, is a challenge that deserves careful considerations and a clear protocol for action (*Figure 7*).

A recent review of trials performed since the 1990s confirmed earlier findings that, in hypertensive patients, lowering systolic BP to <140 mmHg substantially lowered incident HFrEF or HFpEF.[89–91] A large meta-analysis including 123 studies and almost 614 000 patients showed that every 10 mmHg reduction in systolic BP significantly reduced the risk of HF [relative risk or RR: 0.72, 95% confidence interval (CI): 0.67–0.78], supporting lowering systolic BP to <130 mmHg.[91] Previous randomized controlled trials (RCTs) and meta-analyses shown that, compared with a target systolic BP within the 130–139 mmHg range, a systolic BP reduction within the 120–129 mmHg range is accompanied by a significant and sizeable reduction in incident HF among other CV events.[92,93]

The recent statement of the American Heart Association on risk-based primary prevention of HF recommends initiation and treatment goal of BP <130/<80 mmHg in those hypertensive patients at increased risk of CV disease (e.g. patients with HHD).[94] In line with this, the 2025 Joint Scientific Statement from the Heart Failure Society of America and

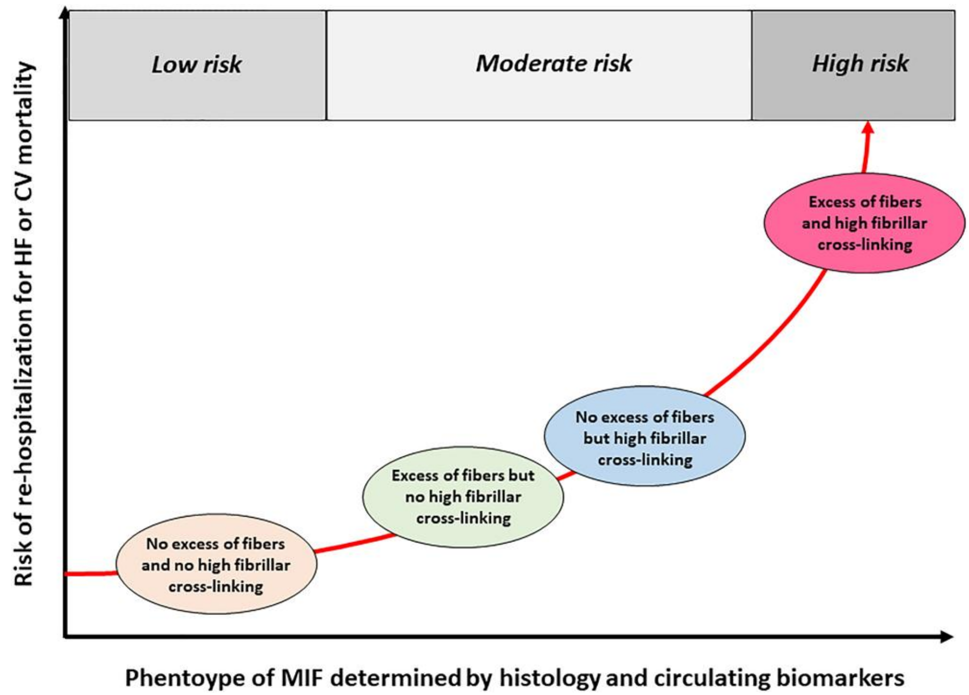


Figure 6 Proposed non-linear relationship between circulating biomarker-based myocardial interstitial fibrosis phenotype and the risk of rehospitalization for heart failure or cardiovascular mortality in hypertensive patients with heart failure. See text for explanations. Prepared from data in reference[88]

The American Society for Preventive Cardiology determined a target BP of <130/80 mmHg for HF prevention.[95] According to the 2023 European Society of Hypertension,[10] the 2024 European Society of Cardiology hypertension guidelines,[11] and the 2025 AHA/ACC guideline of high BP,[12] the optimal recommended treatment goal to prevent HF is 120–129 mmHg for systolic BP and 70–79 mmHg for diastolic BP in most patients, though the European Society of Cardiology do not exclude achieving values of systolic BP under 120 mmHg, if well tolerated, according to the principle of 'lower is better', as recently underlined in a critical comparison.[96] This recommendation is supported by findings from the SPRINT trial showing a 38% reduction in HF incidence and a 23% reduction in all-cause mortality when systolic BP was targeted to <120 mmHg, compared with a target of <140 mmHg.[97]

All major antihypertensive drug classes [i.e. diuretics, including thiazides, chlorthalidone and indapamide, beta-blockers (BBs), angiotensin-converting enzyme inhibitors (ACEIs), angiotensin II receptor blockers (ARBs), and calcium channel blockers (CCBs)] reduce incident HF compared with control groups and thus all may be used for this purpose.[10–12] However, ACEIs and ARBs are more effective than diuretics, BBs and CCBs, confirming the important role of angiotensin II in the development of HF in hypertension.[98–102]

Randomized controlled trials are currently evaluating novel antihypertensive agents such as aldosterone synthase inhibitors (ASIs) or angiotensinogen targeted therapies (Supplementary Table S1), although their efficacy in preventing HF has been scarcely addressed to date. Furthermore, ongoing studies are testing new members of drug classes already commonly used for the treatment of hypertension, methods that may improve compliance with therapies including combination agents containing two to four drug classes, and several nutritional and dietary supplements which can act directly on BP or complement the effects of antihypertensive drugs.[103,104] Overall, additional studies are required to assess the efficacy of these new therapeutic options in reducing the risk of HF caused by hypertension.

While sodium-glucose cotransporter 2 inhibitors and glucagon-like receptor agonists may offer additive or synergistic benefits in reducing BP and CV risk in hypertensive patients in general, patients with co-existing obesity and/or DM and/or CKD or hypertension-mediated organ damage (e.g. HF) may particularly benefit from the use of these drugs.[105]

The role of excess adiposity in hypertension and HF (particularly, HFpEF) has been substantially underappreciated, largely because of reliance on body mass index (rather than waist-to-height ratio) to identify an expanded fat mass. Adiposity causes hypertension, and if excess adiposity is not alleviated, BP lowering with many antihypertensive drugs may not be able to alleviate LV dysfunction.[106] In the current era, therapeutic amelioration of excess adiposity by bariatric surgery or incretin-based drugs (e.g. glucagon-like receptor-1 agonists or dipeptidyl peptidase-4 inhibitors) appears to be more important that only BP lowering to prevent the progression to HF in hypertensive patients with obesity.[106]

Prevention in hypertensive heart disease

A number of studies have demonstrated that on-treatment reversal of LVH with antihypertensive treatment is associated with improved CV events, including lower incident HF risk independent of BP lowering, suggesting LVH itself as a modifiable target for HF prevention. In two meta-analyses of treated hypertensive patients in whom LVH was treated as a categorical variable rather than as a continuous variable, patients with complete regression of LVH (i.e. normalization of LVM) were compared with patients with LVH persistence or new development of LVH.[107,108] These studies documented that compared with persistence or new development of LVH, complete regression of

Table 1 Stages and criteria for clinical diagnosis of hypertensive heart disease

	Patient without HHD	Patient with asymptomatic HHD	Patient with symptomatic HHD
BP elevated without or with treatment or controlled with treatment	Yes	Yes	Yes
No evidence of other heart disease	Yes	Yes	Yes
Signs and symptoms of HF	Absent	Absent	Present
Arrhythmias[a]	Absent	Absent or silent[b]	Present or absent
Abnormalities on cardiac imaging	Absent	Present	Present
	and	and	and
Alterations in circulating biomarkers[c]	Absent	Present or absent	Present

HHD, hypertensive heart disease; BP, blood pressure; HF, heart failure.
[a]Including ventricular and atrial.
[b]Silent arrhythmias (e.g. silent atrial fibrillation) are arrhythmias that occur without any noticeable symptoms, like heart palpitations, difficulty breathing or dizziness.
[c]N-terminal pro-B-type natriuretic peptide and hs-cTnT.

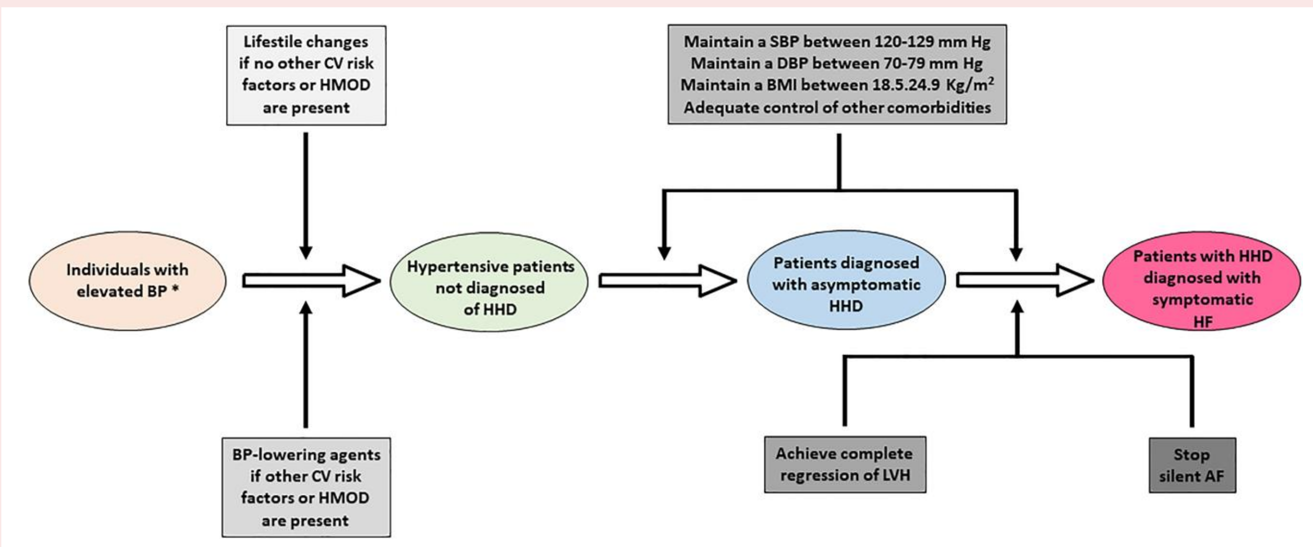


Figure 7 Proposed measures to prevent the risk of heart failure in individuals with elevated blood pressure, patients with hypertension without hypertensive heart disease, and patients with hypertensive heart disease with and without heart failure. CV, cardiovascular; HMOD, hypertension-mediated organ damage; SBP, systolic blood pressure; DBP, diastolic BP; LVH, left ventricular hypertrophy; AF, atrial fibrillation. *As defined in references[10–12]

LVH during antihypertensive treatment is associated with a marked reduction in risk for subsequent CV disease, including HF. This has been confirmed by a meta-regression analysis of 14 trials (all but one including ACEIs or ARBs as treatment) pooling a large number of patients (12 809) and events (2259), that demonstrated that after 1.97 years of follow-up (range 0.50–5 years), LVM reduction without complete regression of LVH did not predict the composite outcome (all-cause death, myocardial infarction, stroke, or new-onset HF) or any of its individual components.[109]

In a large population-based study carried out in 2173 patients with echocardiographic LVH, smaller reductions in both systolic and diastolic BP during follow-up and lower achieved percentage of optimal BP control (<130 mmHg of systolic BP and <80 mmHg of diastolic BP) were associated with less frequent complete regression of LVH.[110] Lack of LVH regression was associated with higher hazard rate of HF and other CV events.[110] Of note, patients who did not achieve complete regression of LVH were older, were commonly women, had a longer duration of hypertension, higher values of LVMI, established hypertension-mediated non-cardiac end organ damage (e.g. carotid atherosclerosis), and higher rates of obesity and DM.[110]

Thus, the impact of antihypertensive treatment in preventing HF in HHD is also related to its ability to completely regress LVH, which, in turn, can be influenced by the patient's overall clinical profile, requiring therapeutic measures that also act on extra-cardiac alterations. In addition, it should be noted that LVH may be very difficult, if not impossible, to completely regress if the abnormality is so consolidated by prolonged exposure, as confirmed by the findings of a population-based study in American Indians.[111] In this conceptual framework, preventing the development of HHD early rather than acting late on LVH would be the strategy of choice to effectively reduce the risk of HF.

Table 2 Uncertainties and gaps, and corresponding potential actions to implement in hypertensive heart disease

Uncertainties and gaps	Corrective actions
1. Unclear clinical definition of HHD as it appears in the International Classification of Diseases	Rule out IHD due to CAD for a more accurate diagnosis of HHD in patients suspected of having it
2. Uncertain knowledge of the correspondence between elevated BP-mediated haemodynamic load and LV growth	Calculate the appropriateness of LV mass in patients with concentric LV growth and diastolic dysfunction
3. Difficulties in understanding the specific impact of hypertension and its comorbidities on the myocardium	Develop research based on molecular imaging in animal models and patients with HHD
4. Limited performance of EKG to identify LVH, differentiate its aetiology and stratify its HF risk	Conduct studies on the ability of AI to cost-effectively overcome these limitations in the real world
5. Lack of information on patients with HHD in whom HF risk classification by imaging techniques is cost-effective	Perform studies to identify patients with HHD and high-risk HF phenotype using imaging techniques
6. Undefined actual added clinical value of circulating biomarkers of heart changes in HHD	Large, long-term 'omic' studies conducted in independent cohorts of patients with HHD are clearly needed
7. An adverse effect on HHD burden is expected due to the actual poor results of hypertension treatment and control	It is necessary to improve population-based primary strategies for effective treatment of hypertension and diagnosis of HHD
8. Suboptimal BP reduction and incomplete resolution of LVH suggest less protection against HF risk from HHD	Treatment of the patient with HHD must meet the established BP and LVMI objectives
9. The risk of HF of HHD is increasing globally, and specially in vulnerable populations due to exposure to various types of factors	It is crucial to prioritize the utilization and allocation of healthcare resources in these vulnerable populations
10. Despite its life-threatening nature, HHD remains unacceptably underdiagnosed	It is urgent to launch scientific and advocacy initiatives to reverse this situation

HHD, hypertensive heart disease; IHD, ischaemic heart disease; CAD, coronary artery disease; BP, blood pressure; LV, left ventricular; EKG, electrocardiogram; LVH, LV hypertrophy; HF, heart failure; AI, artificial intelligence; LVMI, LV mass index.

Several studies demonstrated that myocardial fibrosis present in EMBs from patients with HHD is, at least partially, reversible and that this anti-fibrotic effect is associated with reduction of LV stiffness, LV functional improvement and increase in CFR.[82,112–115] Of note, these studies show that ACEIs and ARBs are effective in reducing collagen deposition,[82,112–115] even compared with other classes of antihypertensive drugs such as BBs or CCBs.[82,112–114] Consistently, chronic treatment with either ACEIs or ARBs was associated with significant decreases in CMR-assessed ECV values in hypertensive rats[113] and patients with HHD,[116] and with serum PICP levels in hypertensive patients[83,117,118] independently of changes in BP. Whether the cardiac anti-fibrotic effect of some antihypertensive drugs is translated into a reduction in the risk of HF is still unknown.

In a large community-based cohort, a hormonal phenotype of primary aldosteronism (i.e. renin suppression and aldosterone excess) was associated with alterations in cardiac structure (including LVH and myocardial fibrosis) and function (including LV diastolic dysfunction and impaired GLS) consistent with HHD.[119] These associations remain significant after adjustment for BP and antihypertensive drugs (including ACEIs and ARBs).[119] In addition, renin suppression and aldosterone excess were associated with incident HF and incident AF, respectively. although these relationships did not remain statistically significant after additional adjustment for hypertension.[119] In line with this observation, the use of the steroidal mineralocorticoid receptor antagonist (MRA) eplerenone induced an effective regression of LVH in hypertensive patients that went beyond what could be expected with BP reduction alone.[120] Therefore, primary aldosteronism screening might be considered namely in patients with uncontrolled treated hypertension,[121] in whom earlier aldosterone-targeting treatment of hypertension with steroidal and non-steroidal MRAs[121] (or with ASIs in the near future) could be considered. Of note, discontinuation of treatment with renin-angiotensin system inhibitors during the 2 weeks previous to the diagnostic tests is needed. However, specifically designed studies are needed to test the benefit of this diagnostic and therapeutic strategy in HHD.

Angiotensin receptor-neprilysin inhibitors (ARNIs) are dual agents comprising a neprilysin inhibitor and an ARB. The ARNI currently in clinical use combines the neprilysin inhibitor sacubitril with the ARB valsartan. In patients with hypertension sacubitril/valsartan demonstrated greater reductions in LVM compared with ARB monotherapy with valsartan[122] or olmesartan.[123] Importantly, the reduction in LVM appears to be independent of changes in BP but is accompanied by a reduction in circulating levels of biomarkers of stressed-injured cardiomyocytes and myocardial fibrosis.[122,123] Recently, sacubitril/valsartan combination was shown to reduce myocardial fibrosis assessed by CMR in hypertensive patients with LVH compared with valsartan alone, despite a similar reduction in BP.[124] This was accompanied by a favourable change in LVM, diastolic function, and circulating NT-proBNP and hs-cTnT.[124] Therefore, the possible advantage of ARNIs over other antihypertensive treatments in HHD, to prevent HF, warrants further investigation.

Several non-randomized sham-controlled clinical studies and some meta-analyses on renal sympathetic denervation (RSD) suggest that RSD might have beneficial effects on LVH, LV diastolic dysfunction, and new-onset/recurrence of AF in patients with resistant hypertension and HHD.[125] Notably, a meta-analysis revealed that the effects of RSD on cardiac alterations were independent of both baseline BP and BP reduction after RSD.[126] It is still necessary to assess the efficacy of RSD in preventing HF in patients with HHD, focusing on parameters regarding sympathetic activity and imaging analyses in the clinical setting, through well designed randomized sham-controlled trials.

Last but not least, lifestyle modifications (i.e. reducing sodium intake, increasing potassium intake, regular physical activity, weight reduction, less than moderate alcohol use, stress management, minimize exposure to noise and air pollution) are also recommended in the treatment of HHD. In a study performed in hypertensive patients it was found that, independently of BP, high dietary sodium may increase LV wall thickness and LVM.[127] Another study in patients with elevated BP found that an increase in the dietary sodium:potassium ratio was associated with higher LVMI, and that sodium consumption was positively associated with atrial diastolic filling fraction.[128] These data suggest that modulating sodium and potassium in the diet may have an important role in the treatment of HHD.

Physical exercise appears to have a positive effect on adverse LV remodelling in hypertensive patients with paradoxical regression or prevention of LVH.[129] This regression appears be independent of changes in BP and body mass with exercise.[129] Experimental data confirm that physical exercise is accompanied by amelioration of histological lesions and molecular alterations present in the hypertensive myocardium.[130,131] Further studies are needed to better understand the response to variable exercise modalities in the hypertensive heart and which populations may benefit most.

Substantial weight loss, particularly in severely obese normotensive or hypertensive individuals, may improve LV structure (e.g. reduction in free wall and septal thickness and internal dimension) and function (e.g. improved LV diastolic filling and ejection fraction), independent of changes in BP.[132,133] Improvements in LVM with weight loss have been reported to occur in hypertensive patients with overweight and are independent of BP reduction.[134]

Uncertainties, gaps, and potential actions to implement

As summarized in *Table 2*, important issues in HHD remain unresolved, the solution of which requires the implementation of multifaceted measures.

(1) The clinical definitions of HHD in the last ICD are vague, as they includes a broader group of alterations related to heart disease caused by direct or indirect effects of hypertension.[135] However, hypertension represents a major contributor to the development of IHD through CAD.[24] The question then arises whether all patients with suspected HHD should undergo assessment of CAD to exclude IHD before making a confident diagnosis of HHD.

(2) Left ventricular growth may exceed individual needs to compensate for the haemodynamic load imposed by increased BP, leading to inappropriate LVM (calculated by the ratio of observed LVM to the value predicted for individual sex, height, and stroke work at rest).[136] Since inappropriate LVM is associated with adverse CV events (including severe HF) more commonly than classically defined LVH,[137] the question is whether its detection should be mandatory in all patients undergoing HHD screening.

(3) Myocardial lesions such as interstitial and perivascular fibrosis and molecular alterations such as impaired mitochondrial metabolism in cardiomyocytes observed in patients with HHD are non-specific as they may be also relevant in DM[138] or CKD.[139] This challenges the mechanistic differentiation between what is intrinsic to HHD and what is the contribution of cardiometabolic comorbidities, which in turn may have therapeutic implications.

(4) A recent study performed in a large series of patients showed that artificial intelligence (AI)-based LVH diagnosis outperformed conventional EKG criteria and was a superior predictor of mortality compared with standard echocardiography-confirmed LVH.[140] Similarly, AI models designed to detect and differentiate LVH on echocardiography, CMR and CT show promising results.[141] However, further studies focusing on their real-life validation and cost-effectiveness analyses are needed before incorporating AI in clinical assessment of HHD.

(5) Objective criteria for identifying patients in whom reclassification of HHD risk by comprehensive echocardiography or CMR is cost-effective are lacking.[142] To improve the cost-effectiveness of risk reclassification based on LVH detection, comprehensive echocardiography should likely be reserved for high-risk hypertensive patients as defined by guidelines.[10–12]

(6) The added value of evaluating a few independent circulating biomarkers in HHD remains undefined, probably due to limitations in study design and analytical methodology. However, findings from recent large and long-term studies using multiprotein/proteomic approaches are promising. For example, the Heart 'Omics' in AGEing (HOMAGE) RCT identified a group of proteins that improved the prediction of incident HF, particularly in patients with hypertension, when added to classical clinical and biochemical risk factors.[143]

(7) Since HHD is inherently exclusive to hypertensive patients (with an assignable risk of 100%), it is obvious that optimal BP control would improve the burden of HHD, something that is far from happening in the real world. Therefore, governments and health organizations should implement and/or improve population-based primary effective diagnosis and treatment strategies for hypertension to prevent HHD in adults and the associated risk of HF. These strategies should take into account that out-of-office BP monitoring (e.g. home BP monitoring or ambulatory BP monitoring) is more closely related to the prevalence of LVH and its response to antihypertensive therapy[144,145] than office BP monitoring.

(8) The ability of antihypertensive drugs to normalize LVMI and thus completely regress LVH is critical for effective prevention of HF and other CV events in HHD. In this regard, there is a lack of randomized studies aimed at defining the most effective and safe target for BP treatment to prevent HHD. In addition, as the effect of BP lowering on LVMI may be influenced by extra-cardiac clinical conditions, treatment of the hypertensive patient with LVH should not only be adequate in terms of BP control, but in many cases should be accompanied by therapeutic measures acting at extra-cardiac levels (e.g. reducing obesity and controlling DM).

(9) Between 1990 and 2021 HHD emerged as one of the two fastest growing contributors to HF not only in low- and middle- but also in high-SDI countries.[2] Reducing the global burden of HHD, requires prioritizing healthcare resource allocation for prevention of hypertension–mediated cardiac organ damage, especially for vulnerable populations exposed to the unfavourable effects of genetic predisposition, race and ethnicity, climate and environmental factors, and social determinants of health.

(10) Despite the growing of HHD, it remains unacceptably underdiagnosed.[54] Therefore, urgent actions are necessary to reverse this trend by launching scientific initiatives (aimed at advancing pathophysiological knowledge and clinical management of HHD) and advocacy measures (aimed at raising awareness of the social and health challenges posed by HHD by itself and within the context of CV-kidney-metabolic health).

Conclusions and perspectives

The growing HF burden of HHD represents an unmet medical need. In fact, HF due to HHD is a major current threat to CV health, and will grow in the future, especially in certain regions of the world and for some socially disadvantaged populations. Despite this, the lack of clinical and structural grading of this multifaceted condition with a continuum from asymptomatic to life-threatening HF must be recognized. In addition, an incomplete mechanistic understanding of HHD, as well as lack of prioritization in clinical management, amplify the threat of consequent HF. Therefore, the time has come to re-emphasize the perception of HHD as a public health challenge worldwide, associated with burdensome long-term consequences for patients and health systems. Within this conceptual framework, adopting an integrative and interdisciplinary approach to HHD research and management is urgent, in order to prevent its increasing risk of HF.

Supplementary data

Supplementary data are available at *European Journal of Heart Failure* online.

Declarations

Disclosure of Interest

A.K. declares potential conflict of interest (COI) in relation to Mineralys. L.J.L. declares potential COI in relation to Arrowhead, AstraZeneca, BaseCure Therapeutics, Kardigan, Lilly, Medtronic, Mineralys, Novartis, Novo Nordisk, Recor, and Ripple Medical. T.B. declares potential COI in relation to AstraZeneca, Daiichi Sankyo, Medtronic, Menarini, Novartis, Novo Nordisk, Roche Diagnostics, Sanofi, and Servier. G.M.R. declares potential COI in relation to Anlylam, AstraZeneca, Bayer, Boehringer Ingelheim, CSL, Medtronic, Menarini, Novartis, Roche, Servier, and Vifor. M.P. declares potential COI in relation to AstraZeneca, Boehringer Ingelheim, and Novo Nordisk. A.M.R. declares potential COI in relation to Abbott Laboratories, Astra Zeneca, Bayer, Boston Scientific, Bristol Myers Squibb, Novo Nordisk, Pfizer, Roche Diagnostics, and Thermo Fisher. B.P. declares potential COI in relation to Anacardio*, Astra Zeneca, Bayer, Boehringer Ingelheim, Cereno Scientific, G3 Pharmaceuticals, KBP Biosciences, Lexicon, Prointel, Sarfez Pharmaceuticals, SC Pharmaceuticals, Sea Star Medical, SQ Innovations, Vifor; he has stock options or equities at Anacardio, Cereno Scientific, G3 Pharmaceuticals, KBO Biosciences, Prointel, Safez Pharmaceuticals, SC Pharmaceuticals, Sea Star Medical, SQ Innovations and Vifor. F.Z. declares potential COI in relation to Alnylam, Bayer, Biopeutics, Boehringer Ingelheim, Cellprothera, Centrix, Centrx Corteria, Cereno, CVRx, Lilly, Lupin, Merck, NovoNordisk, Opalia Recordati, Owkin, Polygon, Ribocure, Roche and Viatris; he has stock options or equities at Cereno pharmaceutical, and he is the founder of Cardiovascular Clinical Trialists (CVCT). M.M. declares potential COI in relation to Abbott Structural Heart, AstraZeneca, Boehringer Ingelheim, Cytokinetics, Edwards Lifesciences, Novo Nordisk, Pharmacosmos, and Roche Diagnostic. M.B. declares potential COI in relation to Abbott, Amgen, AstraZeneca, Bayer, Boehringer Ingelheim, Cytokinetics, Edwards, Medtronic, Novartis, ReCor, Servier and Vifo; he is supported by the Deutsche Forschungsgemeinschaft (German Research Foundation; TTR 219, project number 322900939). J.B. declares potential COI in relation to Abbott, American Regent, Amgen, Applied Therapeutic, AskBio, Astellas, AstraZeneca, Bayer, Boehringer Ingelheim, Boston Scientific, Bristol Myers Squibb, Cardiac Dimension, Cardior, CSL Bearing, CVRx, Cytokinetics, Edwards, Element Science, Faraday, Foundry, Innolife, Impulse Dynamics, Imbria, Inventiva, Ionis, Lexicon, Lilly, LivaNova, Janssen, Medtronic, Merck, Occlutech, Owkin, Novartis, Novo Nordisk, Pfizer, Pharmacosmos, Pharmain, Prolaio, Regeneron, Renibus, Roche, Salamandra, Sanofi, SC Pharma, Secretome, Sequana, SQ Innovation, Tenex, Tricog, Ultromics, Vifor, and Zoll. All other authors report no conflict of interest.

Data Availability

No data were generated or analysed for this manuscript.

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